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ECOMMERCE MANAGEMENT SYSTEM – ERD Diagram

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Course Title: Introduction to Database Systems

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E commerce management system

1. Introduction

This database is designed for an **E commerce management System** where customers can buy products from sellers. The system manages customers, sellers, products, categories, carts, orders, payments, deliveries, reviews, and tax records.
The goal of this database is to store all information safely and maintain proper relationships between different parts of the system.

2. Entities, Attributes, Relationships, Primary Keys & Foreign Keys

Below are all entities with simple explanations.

2.1 Customer

Attributes:

- C_ID
- Name
- Email
- Phone
- Address, City, State, PinCode
- RegistrationDate

Relationships:

- One customer has one cart
- One customer can place many orders
- One customer can write many reviews

Primary Key:

- **C_ID**

Foreign Keys:

- None

2.2 Seller

Attributes:

- S_ID
- Name
- Email
- ContactNo
- BusinessName
- Status
- GL_ID

Relationships:

- One seller has many products
- One seller has many tax records

- One seller is linked to one government institution

Primary Key:

- **S_ID**

Foreign Key:

- **GI_ID** → Government Institution

2.3 Government Institution

Attributes:

- GI_ID
- InstitutionName
- Department
- ContactInfo
- LicenseVerificationPortal
- TaxRate

Relationships:

- One institution manages many sellers
- One institution has many tax records

Primary Key:

- **GI_ID**

Foreign Keys:

- None

2.4 Tax Record

Attributes:

- T_ID
- RecordDate
- TaxAmount
- S_ID
- GI_ID

Relationships:

- Each tax record belongs to one seller
- Each tax record belongs to one government institution

Primary Key:

- **T_ID**

Foreign Keys:

- **S_ID** → Seller
- **GI_ID** → Government Institution

2.5 Category

Attributes:

- CategoryID
- CategoryName
- Description

Relationships:

- One category has many products

Primary Key:

- **CategoryID**

Foreign Keys:

- None

2.6 Product

Attributes:

- P_ID
- Name
- Description
- Price
- StockQuantity
- CategoryID
- S_ID

Relationships:

- One product belongs to one category
- One product belongs to one seller
- One product appears in many cart items
- One product appears in many order details
- One product can have many reviews

Primary Key:

- **P_ID**

Foreign Keys:

- **CategoryID** → Category
- **S_ID** → Seller

2.7 Cart

Attributes:

- cartID
- C_ID
- CreatedAt

Relationships:

- One cart belongs to one customer
- One cart has many cart items
- One cart is used for one order

Primary Key:

- **cartID**

Foreign Key:

- **C_ID** → Customer

2.8 Cart Item

Attributes:

- Ct_ID
- P_ID
- Quantity
- CartID

Relationships:

- One cart item belongs to one cart
- One cart item belongs to one product

Primary Key:

- **Composite PK:** cartID + P_ID

Foreign Keys:

- **Cart_ID** → Cart
- **P_ID** → Product

2.9 Order

Attributes:

- O_ID
- OrderDate
- Status
- TotalAmount
- C_ID

Relationships:

- One order belongs to one customer
- One order has many order details
- One order has one payment
- One order has one delivery

Primary Key:

- **O_ID**

Foreign Keys:

- **C_ID** → Customer

2.10 Order Detail

Attributes:

- OD_ID
- UnitPrice
- Quantity
- Subtotal
- O_ID
- P_ID

Relationships:

- One order detail belongs to one order
- One order detail belongs to one product

Primary Key:

- **OD_ID**

Foreign Keys:
- **O_ID** → Order
- **P_ID** → Product

2.11 Payment
Attributes:
- P_ID
- PaymentMethod
- PaymentDate
- Amount
- PaymentStatus
- O_ID

Relationships:
- One payment belongs to one order

Primary Key:
- **P_ID**

Foreign Key:
- **O_ID** → Order

2.12 Delivery
Attributes:
- D_ID
- DeliveryDate
- DeliveryStatus
- TrackingNumber
- O_ID

Relationships:
- One delivery belongs to one order

Primary Key:
- **D_ID**

Foreign Key:
- **O_ID** → Order

2.13 Review
Attributes:
- R_ID
- ReviewDate
- Rating
- Comment
- C_ID
- P_ID

Relationships:
- One review belongs to one customer

- One review belongs to one product

Primary Key:

- **R_ID**

Foreign Keys:

- **C_ID** → Customer
- **P_ID** → Product

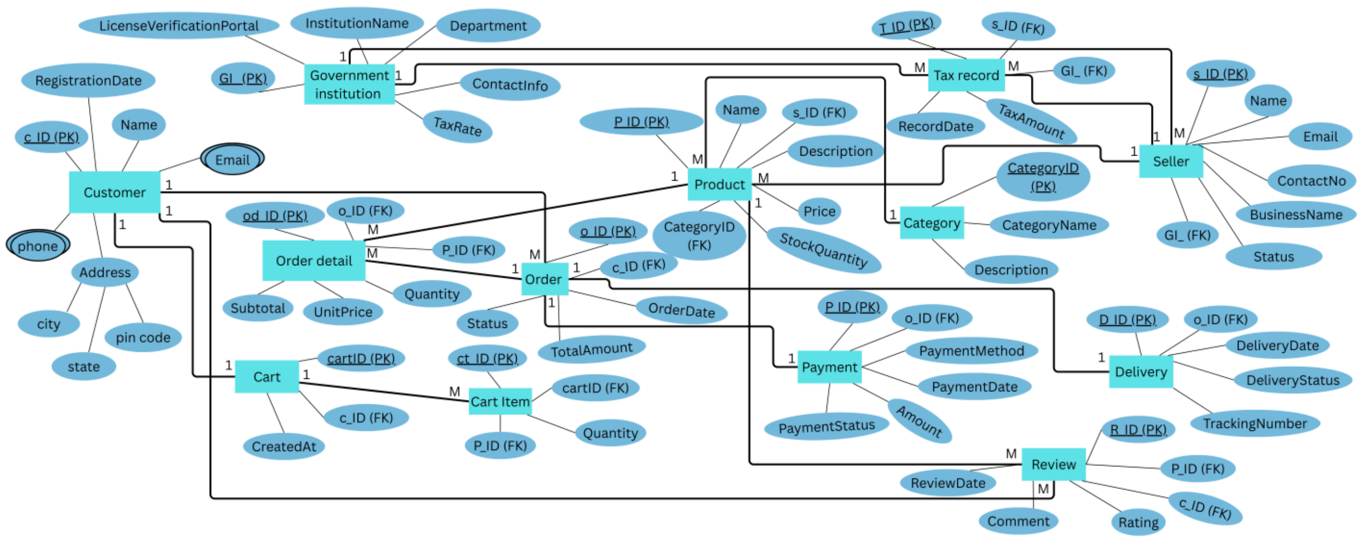
3. Conclusion

This ERD provides a complete database design for a modern e-commerce platform. The system connects customers, sellers, products, orders, payments, deliveries, and reviews in an organized way. It also includes government institutions to monitor taxes, ensuring legal compliance.

The ERD clearly defines all relationships, primary keys, and foreign keys, representing a well-structured relational database model.

This structure supports efficient data storage, retrieval, transaction processing, and secure record management, making it an effective backend system for a real-world e-commerce application.

E commerce management system ERD Diagram :



Complete data base in sql:-

```
CREATE DATABASE ecommerce_db;
USE ecommerce_db;

---------------------Create TABLES-----------------------
---------------------------------------------------------

CREATE TABLE customer (
   C_ID INT PRIMARY KEY AUTO_INCREMENT,
   Name VARCHAR(100),
   Email VARCHAR(100),
   Phone VARCHAR(20),
   Address VARCHAR(255),
   City VARCHAR(50),
   State VARCHAR(50),
   PinCode VARCHAR(10),
   RegistrationDate DATE
);

CREATE TABLE seller (
   S_ID INT PRIMARY KEY AUTO_INCREMENT,
   Name VARCHAR(100),
   Email VARCHAR(100),
   ContactNo VARCHAR(20),
   BusinessName VARCHAR(100),
   Status VARCHAR(50),
   GI_ID INT,
   FOREIGN KEY (GI_ID) REFERENCES government_institution(GI_ID)
);

CREATE TABLE government_institution (
   GI_ID INT PRIMARY KEY ,
   InstitutionName VARCHAR(100),
   Department VARCHAR(100),
   ContactInfo VARCHAR(100),
   LicenseVerificationPortal VARCHAR(255),
   TaxRate DECIMAL(5,2)
);

CREATE TABLE category (
   CategoryID INT PRIMARY KEY AUTO_INCREMENT,
   CategoryName VARCHAR(100),
   Description VARCHAR(255)
);

CREATE TABLE product (
   P_ID INT PRIMARY KEY AUTO_INCREMENT,
   Name VARCHAR(100),
   Description VARCHAR(255),
   Price DECIMAL(10,2),
   StockQuantity INT,
   CategoryID INT,
   S_ID INT,
```

```sql
    FOREIGN KEY (CategoryID) REFERENCES category(CategoryID),
    FOREIGN KEY (S_ID) REFERENCES seller(S_ID)
);

CREATE TABLE cart (
    CartID INT PRIMARY KEY AUTO_INCREMENT,
    C_ID INT,
    CreatedAt DATETIME,
    FOREIGN KEY (C_ID) REFERENCES customer(C_ID)
);

CREATE TABLE cart_item (
    CartID INT,
    P_ID INT,
    Quantity INT,
    PRIMARY KEY (CartID, P_ID),
    FOREIGN KEY (CartID) REFERENCES cart(CartID),
    FOREIGN KEY (P_ID) REFERENCES product(P_ID)
);

CREATE TABLE orders (
    O_ID INT PRIMARY KEY AUTO_INCREMENT,
    OrderDate DATE,
    Status VARCHAR(50),
    TotalAmount DECIMAL(10,2),
    C_ID INT,
    FOREIGN KEY (C_ID) REFERENCES customer(C_ID)
);

CREATE TABLE order_detail (
    OD_ID INT PRIMARY KEY AUTO_INCREMENT,
    O_ID INT,
    P_ID INT,
    UnitPrice DECIMAL(10,2),
    Quantity INT,
    Subtotal DECIMAL(10,2),
    FOREIGN KEY (O_ID) REFERENCES orders(O_ID),
    FOREIGN KEY (P_ID) REFERENCES product(P_ID)
);

CREATE TABLE payment (
    Payment_ID INT PRIMARY KEY AUTO_INCREMENT,
    O_ID INT,
    PaymentMethod VARCHAR(50),
    PaymentDate DATE,
    Amount DECIMAL(10,2),
    PaymentStatus VARCHAR(50),
    FOREIGN KEY (O_ID) REFERENCES orders(O_ID)
);

CREATE TABLE delivery (
    D_ID INT PRIMARY KEY AUTO_INCREMENT,
    O_ID INT,
    DeliveryDate DATE,
    DeliveryStatus VARCHAR(50),
```

```
    TrackingNumber VARCHAR(100),
    FOREIGN KEY (O_ID) REFERENCES orders(O_ID)
);

CREATE TABLE review (
    R_ID INT PRIMARY KEY AUTO_INCREMENT,
    ReviewDate DATE,
    Rating INT,
    Comment VARCHAR(255),
    C_ID INT,
    P_ID INT,
    FOREIGN KEY (C_ID) REFERENCES customer(C_ID),
    FOREIGN KEY (P_ID) REFERENCES product(P_ID)
);

CREATE TABLE tax_record (
    T_ID INT PRIMARY KEY AUTO_INCREMENT,
    RecordDate DATE,
    TaxAmount DECIMAL(10,2),
    S_ID INT,
    GI_ID INT,
    FOREIGN KEY (S_ID) REFERENCES seller(S_ID),
    FOREIGN KEY (GI_ID) REFERENCES government_institution(GI_ID)
);


----------------------------INSERT SAMPLE DATA---------------------------------
------------------------------------------------------------------------------

INSERT INTO customer
(Name, Email, Phone, Address, City, State, PinCode, RegistrationDate) VALUES
('Ahmed', 'ahmed@gmail.com', '03110001111', 'Street 1', 'Islamabad', 'ICT', '44000', '2024-01-10'),
('Sara', 'sara@gmail.com', '03112223333', 'Street 2', 'Lahore', 'Punjab', '54000', '2024-01-12'),
('Bilal', 'bilal@gmail.com', '03113334444', 'Street 3', 'Karachi', 'Sindh', '74000', '2024-01-15');


INSERT INTO seller
(Name, Email, ContactNo, BusinessName, Status, GI_ID) VALUES
('Ali Khan', 'ali@seller.com', '03001112222', 'Ali Store', 'Active', 1),
('Usman Ahmed', 'usman@seller.com', '03002223333', 'Usman Traders', 'Active', 2),
('Hassan Raza', 'hassan@seller.com', '03003334444', 'Hassan Mart', 'Inactive', 3);


INSERT INTO government_institution
(InstitutionName, Department, ContactInfo, LicenseVerificationPortal, TaxRate) VALUES
('FBR', 'Taxation', '051-111111', 'www.fbr.gov.pk', 15.00),
('SECP', 'Corporate', '051-222222', 'www.secp.gov.pk', 12.50),
('PRA', 'Revenue', '042-333333', 'www.pra.punjab.gov.pk', 10.00);


INSERT INTO category (CategoryName, Description) VALUES
('Electronics', 'Electronic items'),
('Clothing', 'Fashion products'),
('Groceries', 'Daily items');
```

```sql
INSERT INTO product
(Name, Description, Price, StockQuantity, CategoryID, S_ID) VALUES
('Laptop', 'HP Laptop', 120000, 10, 1, 1),
('Mobile', 'Samsung Phone', 80000, 15, 1, 2),
('T-Shirt', 'Cotton Shirt', 2500, 50, 2, 3),
('Rice Bag', '5kg Rice', 1200, 100, 3, 1);


INSERT INTO cart (C_ID, CreatedAt) VALUES
(1, '2024-02-01 10:30:00'),
(2, '2024-02-02 11:00:00');


INSERT INTO cart_item (CartID, P_ID, Quantity) VALUES
(1, 1, 1),
(1, 2, 2),
(2, 3, 1);


INSERT INTO orders
(OrderDate, Status, TotalAmount, C_ID) VALUES
('2024-02-05', 'Confirmed', 120000, 1),
('2024-02-06', 'Pending', 80000, 2);



INSERT INTO order_detail
(O_ID, P_ID, UnitPrice, Quantity, Subtotal) VALUES
(1, 1, 120000, 1, 120000),
(2, 2, 80000, 1, 80000);



INSERT INTO payment
(O_ID, PaymentMethod, PaymentDate, Amount, PaymentStatus) VALUES
(1, 'Credit Card', '2024-02-05', 120000, 'Paid'),
(2, 'Cash', '2024-02-06', 80000, 'Pending');


INSERT INTO delivery
(O_ID, DeliveryDate, DeliveryStatus, TrackingNumber) VALUES
(1, '2024-02-07', 'Shipped', 'TRK001'),
(2, '2024-02-08', 'Pending', 'TRK002');


INSERT INTO review
(ReviewDate, Rating, Comment, C_ID, P_ID) VALUES
('2024-02-10', 5, 'Excellent Product', 1, 1),
('2024-02-11', 4, 'Good Quality', 2, 2);


INSERT INTO tax_record
(RecordDate, TaxAmount, S_ID, GI_ID) VALUES
('2024-02-12', 18000, 1, 1),
('2024-02-13', 10000, 2, 2);
```

```sql
------------------------SELECT QUERIES---------------------------
-----------------------------------------------------------------

SELECT * FROM customer;
SELECT Name, Email FROM seller;
SELECT Name, Price FROM product;
SELECT * FROM orders;


------------------------WHERE CLAUSE----------------------------
-----------------------------------------------------------------

SELECT * FROM product WHERE Price > 5000;
SELECT * FROM seller WHERE Status = 'Active';
SELECT * FROM customer WHERE City = 'Lahore';
SELECT * FROM product WHERE StockQuantity < 20;


----------------------AGGREGATION FUNCTIONS---------------------
-----------------------------------------------------------------

----Total Customers
SELECT COUNT(*) AS Total_Customers
FROM customer;

----Total Products
SELECT COUNT(*) AS Total_Products
FROM product;

----Total Stock Value
SELECT SUM(Price * StockQuantity) AS Total_Stock_Value
FROM product;


---------------GROUP BY-------------------

----Products Count Per Category
SELECT c.CategoryName, COUNT(p.P_ID) AS Total_Products
FROM category c
JOIN product p ON c.CategoryID = p.CategoryID
GROUP BY c.CategoryName;



----------------------SORTING AND LIMITING-------------------
-------------------------------------------------------------

----Products Sorted by Price (DESC)
SELECT Name, Price
FROM product
ORDER BY Price DESC;

----Products Sorted by Stock(ASC)
```

```sql
SELECT Name, StockQuantity
FROM product
ORDER BY StockQuantity ASC;


----Top 3 Expensive Products
SELECT Name, Price
FROM product
ORDER BY Price DESC
LIMIT 3;

----Cheapest 2 Products
SELECT Name, Price
FROM product
ORDER BY Price ASC
LIMIT 2;


---------------------------JOINS-------------------------------
----------------------------------------------------------------

-- Customer Orders
SELECT c.Name, o.O_ID, o.TotalAmount
FROM customer c
JOIN orders o ON c.C_ID = o.C_ID;

-- Product Seller
SELECT p.Name, s.BusinessName
FROM product p
JOIN seller s ON p.S_ID = s.S_ID;

-- Product Category
SELECT p.Name, c.CategoryName
FROM product p
JOIN category c ON p.CategoryID = c.CategoryID;

-- Seller Institution
SELECT s.Name, g.InstitutionName
FROM seller s
JOIN government_institution g ON s.GI_ID = g.GI_ID;



---------------------------SUBQUERIES------------------------------
---------------------------------------------------------------------

-- Products above average price
SELECT Name FROM product
WHERE Price > (SELECT AVG(Price) FROM product);

-- Customers who placed orders
SELECT Name FROM customer
WHERE C_ID IN (SELECT C_ID FROM orders);

-- Sellers with products
```

```
SELECT Name FROM seller
WHERE S_ID IN (SELECT S_ID FROM product);

-- Highest priced product
SELECT * FROM product
WHERE Price = (SELECT MAX(Price) FROM product);


---------------------------UPDATE QUERIES--------------------------
-----------------------------------------------------------------
UPDATE product SET Price = 115000 WHERE Name = 'Laptop';
UPDATE seller SET Status = 'Inactive' WHERE S_ID = 2;
UPDATE customer SET City = 'Rawalpindi' WHERE C_ID = 1;
UPDATE product SET StockQuantity = StockQuantity - 2 WHERE P_ID = 1;


---------------------------DELETE QUERIES--------------------------
-----------------------------------------------------------------

DELETE FROM review WHERE Rating < 3;
DELETE FROM product WHERE StockQuantity = 0;
DELETE FROM seller WHERE Status = 'Inactive';
DELETE FROM cart_item WHERE Quantity = 0;


------------------------------VIEWS------------------------------
-----------------------------------------------------------------
CREATE VIEW product_view AS
SELECT Name, Price, StockQuantity FROM product;

CREATE VIEW active_sellers AS
SELECT Name, BusinessName FROM seller WHERE Status = 'Active';

CREATE VIEW customer_city AS
SELECT Name, City FROM customer;

-----------Selection for views---------------

SELECT * FROM product_view;
SELECT * FROM active_sellers;
SELECT * FROM customer_city;


------------------------STORED PROCEDURES------------------------
-----------------------------------------------------------------

:::::::::Get Orders by Customer::::::::

DELIMITER //
CREATE PROCEDURE GetOrdersByCustomer(IN cid INT)
BEGIN
   SELECT * FROM orders WHERE C_ID = cid;
END //
DELIMITER ;
```

```
:::::::::Update Product Stock::::::::::
DELIMITER //
CREATE PROCEDURE UpdateStock(IN pid INT, IN qty INT)
BEGIN
   UPDATE product SET StockQuantity = StockQuantity - qty WHERE P_ID = pid;
END //
DELIMITER ;

:::::::selection functions:::::::::::::
CALL GetOrdersByCustomer(1);
CALL UpdateStock(1, 2);


----------------------Indexes---------------------------------
--------------------------------------------------------------

CREATE INDEX idx_product_price
ON product(Price);

CREATE INDEX idx_customer_city
ON customer(City);

CREATE INDEX idx_order_status
ON orders(Status);

CREATE INDEX idx_payment_status
ON payment(PaymentStatus);

------show index------
SHOW INDEX FROM product;
SHOW INDEX FROM customer;
SHOW INDEX FROM orders;

------drop index------
DROP INDEX idx_customer_city ON customer;
DROP INDEX idx_product_price ON product;


---------------------TRANSACTIONS----------------------
-------------------------------------------------------

START TRANSACTION;

INSERT INTO orders (OrderDate, Status, TotalAmount, C_ID)
VALUES (CURDATE(), 'Pending', 80000, 1);

INSERT INTO order_detail (O_ID, P_ID, UnitPrice, Quantity, Subtotal)
VALUES (LAST_INSERT_ID(), 2, 80000, 1, 80000);

UPDATE product
SET StockQuantity = StockQuantity - 1
WHERE P_ID = 2;

COMMIT;
```

::::::::::::For undo:::::::::::::

```
START TRANSACTION;

INSERT INTO payment (O_ID, PaymentMethod, PaymentDate, Amount, PaymentStatus)
VALUES (1, 'Credit Card', CURDATE(), 80000, 'Failed');

ROLLBACK;
```

Github links :

Muhammad Awais
https://github.com/Awaiskhan0364/IDS-assignment

---THE END---